

Calibration Does Not Compose, Types Destroy Vagueness:

The Hidden-Markov and Fuzzy Primitives Missing from System-One Decision Models

Anh Khoa Doan Ngoc
khoa.doan@edge8.ai

Preprint v1.3 — 1 October 2026

Abstract

System-one decision models—fast, non-generative networks that emit typed, calibrated probabilistic decisions for machine-to-machine pipelines, of which TypeSafe AI’s Jev, trained by Reinforcement Learning for Calibrated Decisions (RLCD), is the announced instance—are audited by hop-level calibration on stationary held-out data. We show that this class omits two structural primitives: recursive belief over a latent state (the hidden-Markov primitive) and graded predicate membership (the fuzzy primitive). We prove two negative results. Lemma 1: when outcomes are coupled through a latent Markov regime, marginal calibration is not preserved under regime shift, trajectory error counts are overdispersed relative to the binomial baseline that hop-level calibration implies, with a closed-form inflation factor, and every permutation-invariant audit statistic—expected calibration error included—has exactly zero power to detect this. Lemma 2: thresholding a typed score makes the composed pipeline action discontinuous; an ε -perturbation of evidence flips a full action on a set of measure $\Theta(\varepsilon)$, every downstream hop then receives an $O(1)$ input change so the conditional flip probability is $\Theta(1)$, and when several hops’ thresholds cross at a common value of a shared latent—as when a harness re-thresholds a score the model has already thresholded—the all-hops-flip event has measure $\Theta(\varepsilon)$ rather than $\Theta(\varepsilon^H)$; a fuzzy membership composed by a t-norm and defuzzified once by a Lipschitz map keeps the action Lipschitz in the evidence. We state the Principle of Deferred Crispification and specify a reference architecture, Belief-State Fuzzy System-One (BSF-S1), whose output object carries a per-regime membership matrix, a calibrated distribution over the degree, and a scaled-likelihood vector over regimes, and collapses exactly once at the actuator at $O(K^2 + KM)$ cost per step. Six CPU-scale experiments, with E1 and E2 run over five seeds against the exact filter as the achievable floor and a Bayes-optimal memoryless head as the strongest memoryless competitor, are consistent with each prediction of the lemmas and expose where a learned approximation falls short of them; two operationally defined audit metrics (trajectory calibration error and action-mass sensitivity) can be run against a closed model from the outside. The paper is architectural: Jev is closed, and nothing here is an empirical claim about its internals.

1 Introduction

1.1 The rise of system-one decision models

For most of the last five years the dominant interface to machine intelligence has been text: a prompt goes in, a string comes out, and software that wants a decision must parse, validate and threshold that string. On 15 September 2026 TypeSafe AI released Jev, which its authors describe as “a frontier-intelligence function call: unstructured state in, typed probabilistic decisions out” (Almeida, 2026). Jev generates no text. Its output schema is fixed in advance through three primitives—*Choice* (an enumeration with a probability attached to each option), *Score* (a rating against ordered levels, returned as a bounded scalar together with a probability

per level) and *Noul*—and Choice and Score answers are accompanied by a confidence score; the documentation’s own example is a ticket-routing Choice over **billing**, **technical** and **account**, returned with a probability per option (TypeSafe AI, 2026). All outputs of a query are produced in a single parallel pass rather than autoregressively, by what the launch post calls a “parallel sampler for maximum efficiency”; the stated end-to-end latency is 70–500 ms, 40 to 200 times faster than frontier models on System-One-shaped queries, input priced at \$0.042 per million tokens and output “too cheap to meter” (Almeida, 2026). The launch drew immediate trade-press coverage (Claburn, 2026); every technical statement in this paper is taken from TypeSafe’s own launch post and documentation. The training method is named Reinforcement Learning for Calibrated Decisions (RLCD) and is said to optimise for “answers with epistemically honest probabilities on System One tasks” (Almeida, 2026).

We take this launch as the concrete instance of a class we call *system-one decision models*: fast, memoryless, typed, calibrated decision heads intended to be consumed by software rather than by people. The class is new enough that its audit vocabulary has been borrowed wholesale from the classification literature. This paper asks whether that vocabulary is adequate.

1.2 The audit practice of the field

The audit practice is hop-level. A model is scored on a held-out sample of (score, outcome) pairs drawn from the same distribution as training; expected calibration error (ECE) or a proper scoring rule summarises the gap between stated and realised frequencies; accuracy benchmarks summarise per-decision correctness. TypeSafe’s documentation states the scope of this measurement with admirable precision: “Calibration is measured across groups of predictions; it does not guarantee that an individual answer is correct” (TypeSafe AI, 2026). This paper takes that sentence seriously and asks a further question: which *groups* is the measurement silent about?

1.3 Problem statement

We identify two omissions.

- (i) **The posterior is memoryless.** The model emits $P(d \mid o)$, calibrated against a frozen marginal prior over the world’s latent state. There is no recursion that carries belief about that state from one decision to the next. When the world’s regime changes, the score remains “calibrated” with respect to a prior that no longer exists.
- (ii) **The output type is crisp.** A typed enumeration commits at design time to sharp boundaries between predicates that, for a positive fraction of inputs, are intrinsically borderline. The model is forced to express ontological vagueness (“this ticket is *somewhat* urgent”) as epistemic uncertainty (“I am 50% sure it is urgent”), and the harness reads the latter as ignorance.

1.4 Thesis

Both omissions are *structural*: they concern what object the model emits and how the harness composes it, not the capacity of the network that produces the object. They survive scaling of a head whose input is the current observation, any improvement in the RL objective, and any amount of additional training data, because they are properties of the type signature of the output. A head fed a window of past observations can approximate the filter, and Section 7 measures how far; but it remains calibrated to a frozen prior, and Lemma 1(iv) applies to it unchanged. They are also *cheap to repair*: the missing machinery is a K -state forward filter and an M -dimensional membership head, costing $O(K^2 + KM)$ operations per step, negligible against a single encoder pass and requiring no generative decode.

1.5 Contributions

- C1** A diagnosis of the temporal hole and Lemma 1 (calibration does not compose): prior-ratio mismatch under regime shift; overdispersion of trajectory error counts with a closed-form inflation factor; and a proof that every permutation-invariant audit statistic is blind to the effect.
- C2** A diagnosis of the vagueness hole and Lemma 2 (cliff cascade): discontinuity of the thresholded pipeline, $\Theta(\varepsilon)$ flip mass with $O(1)$ jumps, $\Theta(1)$ conditional cascade probability, $\Theta(\varepsilon)$ all-hops-flip under coincident thresholds, and Lipschitz continuity of the fuzzy alternative.
- C3** The Principle of Deferred Crispification.
- C4** BSF-S1, a reference architecture with a specified output object, a composition algebra for each component, an exact forward filter over the scaled-likelihood component, and a single actuator collapse.
- C5** Two operationally defined audit metrics, trajectory calibration error (TCE) and action-mass sensitivity (AMS), and six CPU-scale experiments, E1 and E2 run over five seeds against the exact filter as floor, whose results are consistent with every prediction of the lemmas, check equation (3.5) numerically across regime persistence, show the exact filter trajectory-calibrated at every persistence level, show that the temporal hole widens as the memoryless head’s capacity grows, and attribute BSF-S1’s improvement to its transition matrix by ablation.

2 Background and Related Work

2.1 System-one decision models and RLCD

What is public about Jev is the following. The documentation states that a System One model “evaluates a state and returns typed answers and probabilities”, that “their probabilities are optimized against outcomes to reflect uncertainty”, and that such models “do not write replies, produce code, or generate explanations of their reasoning” (TypeSafe AI, 2026); both the documentation and the launch post state that all questions in a request are evaluated in parallel (TypeSafe AI, 2026; Almeida, 2026). Two further statements in the launch material are load-bearing for this paper. First, the recommended integration pattern is to “ask independent questions together ... combine the answers with deterministic checks in code, then route the case for action or review” (TypeSafe AI, 2026): the harness thresholds the returned probabilities. Second, the authors observe that “the most reliable real-world workflows tend to have many independent, decomposed questions, with fine-grained behavior that’s dependent on probabilities instead of discrete decisions. The end result is discrete branching [...]” (Almeida, 2026). This is an accurate description of the pipeline whose composition properties Sections 3 and 4 analyse. The launch material also claims consistency—the model “returns similar answers for similar inputs”—which is a Lipschitz-type property of the *score*; Lemma 2 concerns what happens to that property once the score is thresholded.

Jev’s weights, architecture and training data are closed. Every statement in this paper is about the *class* of memoryless typed calibrated heads, of which Jev is the announced instance; nothing here is an empirical claim about Jev’s internals.

2.2 Hidden Markov models and Bayesian filtering

Let $s_t \in \{1, \dots, K\}$ be a latent regime following a Markov chain with transition matrix $A \in \mathbb{R}^{K \times K}$, $A_{ij} = P(s_{t+1} = j \mid s_t = i)$, and stationary distribution π with $\pi^\top A = \pi^\top$. Evidence o_t is

emitted with $P(o_t | s_t)$. The forward filter (Rabiner, 1989)

$$b_t(s) \propto P(o_t | s) \sum_{s'} A_{s's} b_{t-1}(s') \quad (2.1)$$

is exact inference given A and the emission model. Regime-switching econometrics (Hamilton, 1989) is the same recursion with Gaussian autoregressive emissions, run over the augmented state of current and lagged regimes; partially observable Markov decision processes (Kaelbling et al., 1998) use the same belief update with action-conditioned transitions. The hybrid HMM/neural-network literature (Bourlard and Morgan, 1994) established that a discriminatively trained network emitting $P(s | o)$ can serve as an emission model once its outputs are divided “by the prior probabilities of the classes observed on the training set” (p. 128), giving scaled likelihoods $P(o | s)/P(o) = P(s | o)/\pi(s)$. This is the bridge used in Section 3.5.

2.3 Fuzzy sets and fuzzy logic

A type-1 fuzzy set on X is a membership function $\mu : X \rightarrow [0, 1]$ (Zadeh, 1965). Conjunction and disjunction are *truth-functional*: the degree of “ A and B ” is $\mathsf{T}(\mu_A, \mu_B)$ for a t-norm T (minimum, product, Łukasiewicz), and the degree of “ A or B ” is a t-conorm S (Klement et al., 2000). Minimum is the unique idempotent t-norm (Klement et al., 2000). Mamdani control (Mamdani and Assilian, 1975) evaluates a rule base, aggregates the consequents into an output membership, and selects one action *once*, at the plant input; the original controller took the action of largest membership grade (the midpoint of a plateau), and centroid defuzzification became common later. A type-2 fuzzy set (Zadeh, 1975; Mendel, 2007) attaches to each x a fuzzy set of admissible membership grades, whose secondary grades are possibilities rather than probabilities, and so separates “how much x is M ” from “how sure we are about how much.”

2.4 Calibration and proper scoring

A score f is marginally calibrated under P if $\mathbb{E}_P[Y | f(X) = p] = p$ for all p . ECE bins predictions by confidence and takes the bin-size-weighted average of the gap between accuracy and confidence (Guo et al., 2017); for a binary score it is the binned L^1 gap between $\mathbb{E}[Y | f(X)]$ and $f(X)$. A scoring rule is strictly proper if the true distribution uniquely optimises the expected score (Gneiting and Raftery, 2007), who maximise it in their positive orientation; we use the equivalent loss orientation, in which it is minimised. Calibration of a marginal says nothing about the joint distribution of several decisions; that distinction is the entire content of Section 3.

2.5 Calibration over time, under shift, and on subgroups

Three literatures bear directly on Section 3, and each is a partial answer to it. *Sequential calibration* is as old as the notion itself: Dawid (1982) formalised calibration for sequences of probability forecasts and showed that a coherent Bayesian expects to be well calibrated, and Foster and Vohra (1998) showed that a randomised forecaster can be ε -calibrated, with probability close to one, against any outcome sequence chosen adaptively by an adversary who cannot see the forecaster’s current randomisation. Foster–Vohra calibration, like ECE, is a function of the empirical distribution of (forecast, outcome) pairs, with forecasts on a fixed finite grid, and so at any fixed horizon shares the blindness of Lemma 1(iv). Dawid’s basic notion likewise conditions only on the forecast value, but his calibration theorem extends it to any *admissible* selection process, a rule whose decision to include trial t uses only information available before outcome t . A rule such as “select t whenever $E_{t-1} = 1$ ” is admissible and would expose clustering; Dawid himself selects trials following an occurrence to test a Bernoulli model against Markov dependence. TCE can be read as a window-count instance of that idea, and

history-dependent selection is exactly what hop-level audit practice omits. *Prior shift* on the latent regime is the static version of Lemma 1(i). Saerens et al. (2002) showed that when class priors change, a classifier’s posteriors can be reweighted by the ratio of new to old priors, with the new priors estimated by EM on unlabelled deployment data; Lipton et al. (2018) detect and correct label shift with a black-box predictor’s confusion matrix; and Ovadia et al. (2019) document that calibration measured in distribution does not survive dataset shift. The forward filter of Section 3.5 is the dynamic generalisation of the Saerens reweighting: the scaled likelihood is exactly the posterior divided by the old prior, and the predicted belief $A^\top b_{t-1}$ plays the role of the new prior, propagated by A every step instead of estimated once; the analogue of their EM step is Baum–Welch for A . Note that the shift is in the latent s , not in the decision target d : on (o, d) both $P(o)$ and $P(d | o)$ change and $P(o | d)$ is not invariant, so label-shift correction applied to d would be misspecified. Online recalibration (Gibbs and Candès, 2021) adapts a single global map from realised outcomes with a long-run guarantee. The online-Platt baseline of Section 7.3 is of this type: it repairs the shifted marginal but not the sojourn. The history-window head is a different repair: it tracks the sojourn closely but remains tied to the training prior. *Subgroup calibration* (Hébert-Johnson et al., 2018) requires calibration on every subpopulation identifiable by a class of functions of the input. The subpopulation that matters here, $\{t : s_t = k\}$, is not a function of any observable, current or past. When the input is o_t , the Bayes-optimal memoryless head is already calibrated conditional on o_t itself under P_{tr} , and hence multicalibrated with respect to every function of o_t ; yet it is the worst row of Table 2. Multicalibration over functions of the current input therefore cannot separate the frozen-prior head from the filter; over functions of the history it can at best enforce calibration on groups weighted by $P(s_t = k | o_{1:t})$, which the filter satisfies by construction; and it is a property under P_{tr} that says nothing under shift. This is the precise sense in which the groups of Section 1 are “silent”. Post-hoc calibrators (Guo et al., 2017; Kull et al., 2019) are hop-level repairs and inherit the same limitation.

2.6 Scope and non-goals

We make no claims about generative models, network architecture, scale, context length or caching; no claims about cognition (the “system-one” label follows Kahneman’s popularisation (Kahneman, 2011) of the System 1 / System 2 terms, which he attributes to Stanovich and West, and is used as a label only). We study the decision *output object* and its *composition* across hops.

3 Analysis I: The Temporal Hole

3.1 The memoryless calibrated posterior

Let the world have latent regime s_t as in Section 2, evidence o_t with $P(o_t | s_t)$, and decision target $d_t \in D$ with $P(d_t | o_t, s_t)$. The system-one head is $f(o) \approx P_{\text{tr}}(d | o)$, where the training distribution mixes regimes with their stationary weights:

$$P_{\text{tr}}(d | o) = \sum_s P(d | o, s) P_{\text{tr}}(s | o), \quad P_{\text{tr}}(s | o) \propto P(o | s) \pi(s). \quad (3.1)$$

An RLCD-style objective can certify that f is calibrated under P_{tr} . It cannot certify anything about $P(d | o)$ under a distribution in which the regime weights differ, because the objective never observes s and never observes more than one time step at once.

3.2 Regime shift as prior death

Suppose that at serving time the regime marginal is $\pi' \neq \pi$: a sojourn in a rare regime, or a permanent drift. The true conditional becomes

$$P'(d | o) = \sum_s P(d | o, s) P'(s | o), \quad P'(s | o) \propto P(o | s) \pi'(s). \quad (3.2)$$

The head still outputs (3.1). Its score is “correct” with respect to a prior that has died. A held-out set drawn from P_{tr} has calibration error identically zero for a perfectly trained f , so the audit is structurally blind to (3.2). This is not a failure of training; it is the absence of a variable.

3.3 Lemma 1

Lemma 1 (Calibration does not compose). *Let $d \in \{0, 1\}$ (the K -ary case is identical componentwise). Assume the chain is stationary ($s_1 \sim \pi$) and that, conditional on the state path $s_{1:T}$, the pairs (o_t, d_t) are mutually independent, with (o_t, d_t) depending on the path only through s_t . Let f be perfectly calibrated under P_{tr} . Let $E_t = \mathbf{1}\{\text{decision at } t \text{ is wrong}\}$, $e_s = P(E_t = 1 | s_t = s)$ the regime-conditional error rate of the fixed decision rule, and $\bar{e} = \sum_s \pi(s) e_s$.*

(i) (Prior-ratio mismatch.) *Under serving marginal π' , the pointwise miscalibration of f is*

$$\gamma(o) := P'(d = 1 | o) - f(o) = \sum_s P(d = 1 | o, s) [P'(s | o) - P_{\text{tr}}(s | o)], \quad (3.3)$$

nonzero whenever the regime-conditional answers differ across s and $\pi' \neq \pi$. The ECE under P' is bounded below by $|\mathbb{E}_{P'} \gamma(o)|$ while the ECE under P_{tr} is zero.

(ii) (Overdispersion.) *For a two-regime chain $A = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ with $\lambda := 1 - a - b \in (-1, 1)$,*

$$\text{Cov}(E_t, E_{t+k}) = \lambda^k \text{Var}_{\pi}(e), \quad \text{Var}_{\pi}(e) := \sum_s \pi(s) e_s^2 - \bar{e}^2. \quad (3.4)$$

Hence the window error count $N_T = \sum_{t=1}^T E_t$ satisfies

$$\text{Var}(N_T) = T \bar{e}(1 - \bar{e}) + 2 \text{Var}_{\pi}(e) \sum_{k=1}^{T-1} (T - k) \lambda^k, \quad (3.5)$$

and the asymptotic variance-inflation factor over the binomial baseline implied by hop-level calibration is

$$\varphi := \lim_{T \rightarrow \infty} \frac{\text{Var}(N_T)}{T \bar{e}(1 - \bar{e})} = 1 + \frac{2\lambda}{1 - \lambda} \cdot \frac{\text{Var}_{\pi}(e)}{\bar{e}(1 - \bar{e})}. \quad (3.6)$$

$\varphi > 1$ iff $\lambda > 0$ and $\text{Var}_{\pi}(e) > 0$. Error clustering has the correlation length of the regime sojourns, which are geometric with means $1/a$ and $1/b$. For a K -regime chain with spectral decomposition $A^k = \mathbf{1}\pi^\top + \sum_{i \geq 2} \lambda_i^k P_i$, (3.4) becomes $\text{Cov}(E_t, E_{t+k}) = \sum_{i \geq 2} \lambda_i^k e^\top \text{diag}(\pi) P_i e$.

(iii) (All-correct probability.) *$P(N_T = 0) = \mathbb{E}[\prod_t (1 - e_{s_t})] \geq (1 - \bar{e})^T$ whenever $\lambda \geq 0$, strictly iff $\lambda > 0$ and $\text{Var}_{\pi}(e) > 0$. The trajectory distribution is heavier in both tails than the independent baseline.*

(iv) (Permutation-invariance blindness.) *ECE, any proper score averaged over hops, and any statistic that is a function of the empirical distribution of $(f(o_t), y_t)$ pairs is invariant under permutation of t . A clustered sequence and its random shuffle have identical such statistics while (3.5) differs. Hop-level audits therefore have exactly zero power against (ii)–(iii).*

Proof sketch. (i) is (3.1) subtracted from (3.2). (ii): condition on (s_t, s_{t+k}) ; by the independences $\mathbb{E}[E_t E_{t+k}] = e^\top \text{diag}(\pi) A^k e$; substitute $A^k = \mathbf{1}\pi^\top + \lambda^k(I - \mathbf{1}\pi^\top)$, valid for every 2×2 stochastic matrix; the first term gives $\bar{e}^2$ and the second $\lambda^k \text{Var}_\pi(e)$. Summing covariances gives (3.5); the geometric series gives (3.6). (iii): $x \mapsto x^T$ is convex on $[0, 1]$; at $\lambda = 1$ Jensen gives the inequality directly, and for $0 < \lambda < 1$ the two-state chain is the mixture $A = (1 - \lambda)\mathbf{1}\pi^\top + \lambda I$, whose paths consist of independent frozen blocks. (iv): ECE is a function of the multiset of pairs, and multisets are permutation-invariant. Full proofs are in Appendix B. $\square$

Numerical check. Section 7.8 verifies (3.5) by simulation across $\lambda \in [0, 0.98]$, (3.4) at $\lambda = 0.95$, and the closed forms of (ii)–(iii) at a fixed operating point.

3.4 Consequence

Risk in a pipeline is a trajectory-level object. The quantity a downstream consumer cares about is the distribution of N_T —how many wrong decisions arrive in this window, and whether they arrive together—not $\bar{e}$. Hop-level calibration certifies $\bar{e}$ and is silent on φ . Two models with identical ECE can differ in φ by an arbitrary factor, and a consumer that budgets for binomial error is under-provisioned by exactly that factor.

3.5 Repair primitive

Treat the network as an emission model and run the exact filter.

1. Train a regime head $g(o) \approx P_{\text{tr}}(s \mid o)$ (K -way softmax) alongside per-regime decision heads $h_s(o) \approx P(d \mid o, s)$.
2. Convert to a scaled likelihood: $\Delta b_t(s) := g_s(o_t)/\pi(s)$, normalised to sum to one, where π is the regime marginal of the data on which g was trained (the stationary distribution of A when the training stream is stationary). The omitted normaliser is a function of o_t alone and cancels in (3.7). Dividing by any other prior—for instance the stationary distribution of a later re-estimate of A —silently mis-scales the evidence; our first implementation made exactly this error.

3. Filter:

$$\tilde{b}_t(s) = \sum_{s'} A_{s's} b_{t-1}(s'), \quad b_t(s) = \frac{\tilde{b}_t(s) \Delta b_t(s)}{\sum_{s''} \tilde{b}_t(s'') \Delta b_t(s'')}. \quad (3.7)$$

4. Decide:

$$P(d_t \mid o_{1:t}) = \sum_s h_s(o_t) b_t(s). \quad (3.8)$$

Given a correct (A, g, h) the decision posterior is the true conditional, hence calibrated on any window under any π' , and its implied joint over N_T is the true joint: trajectory calibration is a consequence of exact inference, not an additional objective. The reason the carried object is a scaled likelihood and not a posterior is that two posteriors $P(s \mid o_1)$ and $P(s \mid o_2)$ each already contain π ; multiplying them counts π twice. Dividing each by $\pi(s)$ leaves objects proportional to $P(o_i \mid s)$, which do multiply under conditional independence.

4 Analysis II: The Vagueness Hole

4.1 Two distinct “not-sures”

Epistemic uncertainty: the world is in exactly one crisp state, we do not know which, and a probability over states sums to one. Ontological vagueness: the predicate itself has no sharp

boundary; membership is a degree; degrees over different predicates need not sum to anything. A single number in $[0, 1]$ attached to an enumeration value conflates the two, and every consumer of that number must guess which one it is receiving. Neither is the aleatoric uncertainty of the ML literature (Hüllermeier and Waegeman, 2021), which is still uncertainty about a crisp outcome.

4.2 Typed enumerations as frozen ontology commitments

A schema $\text{priority} \in \{\text{low}, \text{medium}, \text{high}\}$ commits at design time to sharp boundaries the data do not have. An input that is genuinely borderline between medium and high is best described by memberships $(0, 0.5, 0.5)$. A calibrated probabilistic head trained on inconsistent labels for such inputs learns $p \approx (0, 0.5, 0.5)$ as a probability, which the harness reads as “the model does not know” and routes to escalation, retry, or a more expensive model. The information “this is a borderline case and will remain one no matter how many times you ask” is destroyed, and the cost of that destruction is paid on every borderline input forever. The Score primitive in Jev’s schema (TypeSafe AI, 2026) returns a bounded scalar together with a probability per ordered level: the scalar expresses a degree and the per-level distribution expresses uncertainty about *which level*, but both are tied to one fixed discretisation, and neither separates the degree from a calibrated distribution over the degree itself, which is the type-2 distinction of Section 4.5.

4.3 Lemma 2

Lemma 2 (Cliff cascade). *Let evidence $x \in X \subset \mathbb{R}^n$ have distribution P_X . There are H hops; hop h has an L -Lipschitz score $f_h : X \rightarrow [0, 1]$ and threshold $\tau_h \in (0, 1)$. Assume $f_h \in C^1$, that the pushforward of P_X under f_h has a density p_h positive and finite at τ_h , and that $\|\nabla f_h\| \geq c > 0$ on an open neighbourhood of $f_h^{-1}(\tau_h)$.*

Crisp pipeline: $a_h(x) = \mathbf{1}\{f_h(x) \geq \tau_h\}$ and outcome $G(x) = \bigwedge_h a_h(x)$ (any non-constant Boolean combination gives the same conclusions). Sequential crisp pipeline: hop $h+1$ ’s evidence includes a_h , $x_{h+1} = (x, a_h)$. Fuzzy pipeline: $\mu_h = f_h$ read as a membership, composed by a t -norm T that is 1-Lipschitz in ℓ^1 on $[0, 1]^H$ (minimum and product both are), and defuzzified once by an L_D -Lipschitz map D ; action $u(x) = D(\mathsf{T}(\mu_1(x), \dots, \mu_H(x)))$.

- (i) (Discontinuity vs. Lipschitz.) *G is discontinuous on $\bigcup_h (f_h^{-1}(\tau_h) \cap \{a_j(x) = 1 \ \forall j \neq h\})$, the part of each threshold set on which the remaining gates pass; assume this set has positive P_X -density, so that its ε -neighbourhood has positive measure for every $\varepsilon > 0$. The fuzzy action satisfies, for all x, x' ,*

$$|u(x) - u(x')| \leq L_D \sum_h |\mu_h(x) - \mu_h(x')| \leq L_D H L \|x - x'\|. \quad (4.1)$$

- (ii) ($\Theta(\varepsilon)$ flip mass with $O(1)$ jumps.) *Let $F_\varepsilon = \{x : \exists x', \|x' - x\| \leq \varepsilon, G(x') \neq G(x)\}$. Then $P_X(F_\varepsilon) = \Theta(\varepsilon)$, with per-hop coefficient between $c p_h(\tau_h) \rho_h$ and $2L p_h(\tau_h)$, where $\rho_h > 0$ is the conditional probability that the remaining gates pass given $f_h = \tau_h$. On F_ε the worst-case action change is 1. For the fuzzy pipeline the worst-case action change is at most $L_D H L \varepsilon$ at every x ; there is no set on which it is $O(1)$.*

- (iii) (Cascade.) *In the sequential crisp pipeline, suppose hop $h+1$ actually uses hop h ’s output, $f_{h+1}(x, 1) - f_{h+1}(x, 0) \geq c' > 0$ for all x , and that the conditional law of $f_{h+1}(x, 0)$ given $f_h(x) = \tau_h$ has a density bounded below on $[\tau_{h+1} - c', \tau_{h+1} + c']$. Conditional on a_h flipping, hop $h+1$ ’s score moves by at least c' in a fixed direction, independent of ε , so*

$$\begin{aligned} q_{h+1} &:= \inf_{\varepsilon > 0} P(a_{h+1} \text{ flips} \mid a_h \text{ flips}) \\ &\geq \inf_{\varepsilon > 0} P_X(\tau_{h+1} - c' \leq f_{h+1}(x, 0) < \tau_{h+1} \mid |f_h(x) - \tau_h| \leq \varepsilon) = \Theta(1). \end{aligned} \quad (4.2)$$

Conditional on the first hop flipping, the expected number of further flips is at least $\sum_{h=2}^H \prod_{j=2}^h q_j$, which does not vanish as $\varepsilon \rightarrow 0$. In the sequential fuzzy pipeline hop $h+1$ receives μ_h in place of a_h ; if each f_{h+1} is L -Lipschitz in its full input, the recursion gives $|\Delta\mu_h| \leq (\sum_{j \leq h} L^j)\varepsilon$ and a total action change of at most $L_D H \sum_{h=1}^H L^h \varepsilon$: geometric in L , still $O(\varepsilon)$.

(iv) (Common-mode coupling.) Suppose the hops’ scores are monotone functions of a shared latent scalar z , $f_h(x) = \phi_h(z, \xi_h)$, and the perturbation acts on z . If the hops’ crossing points coincide—there is a z^* with $\phi_h(z^*, \xi_h) = \tau_h$ for every h , as when several hops threshold the same score (a model threshold followed by a harness threshold) or evaluate the same regime-dependent predicate—then $P(\text{all } H \text{ hops flip}) = \Theta(\varepsilon)$. If instead the crossing points are spread with a continuous joint density, $P(\text{all flip}) = \Theta(\varepsilon^H)$ even under a common-mode perturbation, the same order as under independent per-hop perturbations. Coupling therefore bites through coincident thresholds, which is precisely the double-collapse pattern of Section 5.

Proof sketch. (i) A step function composed with a continuous map is discontinuous on the preimage of the threshold; (4.1) chains Lipschitz bounds using $|\prod_h a_h - \prod_h b_h| \leq \sum_h |a_h - b_h|$ on $[0, 1]^H$ (telescoping) and $|\min a - \min b| \leq \max_h |a_h - b_h|$. (ii) F_ε contains the band $\{\tau_h - c\varepsilon \leq f_h < \tau_h\}$, reachable by moving ε along ∇f_h , and is contained in $\{|f_h - \tau_h| \leq L\varepsilon \text{ for some } h\}$ by Lipschitzness; both have mass $\Theta(\varepsilon)$ by the density assumption. (iii) is (4.2). (iv) With coincident crossing points the set of z within ε of z^* has measure $\Theta(\varepsilon)$ and monotonicity forces simultaneous crossing; with spread crossing points all H of them must lie within a 2ε window that z then enters, an event of order $\varepsilon^{H-1} \cdot \varepsilon$. Appendix B. $\square$

Remark 1 (On the defuzzifier). Inequality (4.1) requires D Lipschitz. The centroid $D(\mu_{\text{out}}) = \int y \mu_{\text{out}}(y) dy / \int \mu_{\text{out}}(y) dy$ is Lipschitz in μ_{out} (sup norm) on the set $\{\int \mu_{\text{out}} \geq m_0 > 0\}$; the architecture must enforce the floor m_0 (a regularised centroid). Mean-of-maxima and height defuzzifiers are *not* continuous and would reintroduce the cliff. The principle of Section 5 therefore names the defuzzifier.

Remark 2 (What the lemma does not claim). For a smooth score distribution, the *expected* L^1 sensitivity of the first hop is $O(\varepsilon)$ in both pipelines. The separation is in the per-point worst case (a jump of 1 versus a bound of $L\varepsilon$), in the cascade (a $\Theta(1)$ conditional flip probability versus $O(\varepsilon)$ propagation), and in coincident thresholds ($\Theta(\varepsilon)$ versus $\Theta(\varepsilon^H)$). Readers who have seen “fuzzy is smoother” asserted without qualification should note that (iii) and (iv) are where the content lies.

4.4 The wrong algebra

Probability is not truth-functional: $P(A \wedge B)$ is not determined by $P(A)$ and $P(B)$ without an independence assumption. A t-norm is truth-functional by construction: $T(\mu_A, \mu_B)$ is the degree of “ A and B .” When a harness multiplies typed probabilities across hops it silently asserts independence of predicates that are typically about the same object; when it takes a maximum it asserts nothing coherent. The product t-norm coincides numerically with independent multiplication but is licensed by a different semantics. Minimum is the unique idempotent t-norm (Klement et al., 2000), and idempotence is exactly the property that prevents a predicate from being double-counted when it enters two rules: product gives μ^2 , which is the origin of the “confident by repetition” failure in which asking the same question twice manufactures certainty.

4.5 Repair primitive

Emit a membership vector $\mu(x) \in [0, 1]^M$ over the M linguistic predicates the schema cares about, with no sum constraint, and a type-2 layer $Q[\mu]$ —a distribution over $[0, 1]^M$, in practice

per-predicate Beta parameters—carrying calibrated uncertainty about the degree. Q is a probabilistic analogue of a type-2 fuzzy set: the secondary grades of Zadeh (1975) and Mendel (2007) are possibilities, whereas we use a probability distribution so that Q can be audited with proper scores. Calibration of Q is quantile calibration, $P(\mu_{\text{true},m} \leq Q_m^{-1}(\alpha)) = \alpha$, audited by the continuous ranked probability score or the pinball loss against elicited degrees. The two axes are never merged.

5 The Principle of Deferred Crispification

5.1 Three structures, three algebras

Structure	Object	Composition algebra	Audit
Vagueness	$\mu \in [0, 1]^M$	t-norm / t-conorm, rule base	agreement with elicited degrees
Temporal coupling	$\Delta b \in \mathcal{S}^K$ (scaled likelihood)	pointwise product with predicted belief	trajectory calibration (TCE)
Uncertainty about degree	$Q[\mu]$	mixture / posterior update	proper scores (CRPS, pinball)

Table 1: Each structure has its own object, its own composition rule and its own audit. Collapsing any of them early substitutes the wrong algebra for the right one.

5.2 Statement

Principle 1 (Deferred Crispification). No structure may be collapsed before the actuator. Collapse—defuzzify, threshold, sample—happens exactly once, with full decision context: the belief over regimes, the cost matrix, and every membership in hand.

Each early collapse is a specific information loss. Thresholding μ discards the degree (Lemma 2). Marginalising b_t discards the regime (Lemma 1). Collapsing Q to its mean discards whether the degree is known. None of these losses can be recovered downstream, because the downstream consumer receives only the collapsed object.

5.3 Output object

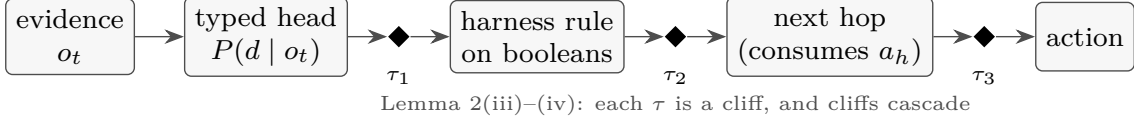
$$O(x_t) = (\mu_t \in [0, 1]^{K \times M}, \quad Q_t, \quad \Delta b_t \in \mathcal{S}^K). \quad (5.1)$$

μ_t is emitted *per regime*: row s is the membership vector the model would report if the regime were s . Q_t likewise. This is the fully deferred form; mixing over regimes is itself a collapse and is left to the actuator. When $K \times M$ is too large, the harness may pass the belief-weighted mixture $\bar{\mu}_t = \sum_s b_t(s) \mu_t^{(s)}$ —a convex combination, hence still a membership vector—at the cost of an early collapse over s , which must be declared.

5.4 Interface contract for harnesses

Downstream hops consume O uncollapsed. Rule bases compose μ with t-norms and t-conorms and never threshold. Filters compose Δb with A by (3.7) and never receive a posterior from another hop, because a posterior would double-count π . Scorers audit Q with proper scores and audit the filter with TCE. Only the actuator collapses.

(a) *Typed pipeline: collapse at every hop*



(b) *BSF-S1: carry the structure, collapse once*

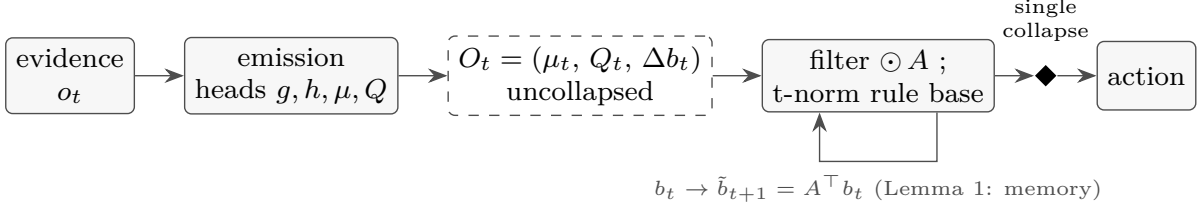


Figure 1: The two pipelines of this paper. (a) A typed pipeline collapses at every hop: the model’s score is thresholded, the harness combines booleans and thresholds again, and a downstream hop consumes the boolean. Each diamond is a discontinuity (Lemma 2(i)); a flip at one diamond moves the next hop’s input by $O(1)$ (part (iii)); if two diamonds threshold the same score, as when a harness re-thresholds a score the model has already thresholded, they flip together (part (iv)). (b) BSF-S1 passes the uncollapsed object O_t forward, composes each component with its own algebra, carries the belief across time (Lemma 1), and collapses once.

5.5 Precedent audit

Four independent engineering traditions arrived at deferred crispification in their own dialects, and their agreement is the strongest evidence we can offer for the principle short of the experiments of Section 7.

Fuzzy control. Mamdani’s steam-engine controller (Mamdani and Assilian, 1975) set the pattern: the sampled plant state enters every rule, each rule fires to a degree, the consequents are aggregated into one output membership, and a single deterministic action is chosen at the plant input. Lemma 2 gives a reason to prefer that ordering over thresholding inside the rule base, which would place a cliff at every internal threshold. It also sharpens it: the original controller chose the action of largest membership grade, which is the mean-of-maxima rule the Remark after Lemma 2 shows to be discontinuous. The principle fixes *where* the collapse happens; the Remark fixes *which* collapse keeps the action Lipschitz.

Probabilistic forecasting. Probabilistic forecasting has become routine in weather and climate prediction and in computational finance, and proper scoring rules arise naturally from decision problems (Gneiting and Raftery, 2007); a full predictive distribution serves consumers with different cost–loss ratios, which a single point forecast cannot. The distribution is carried to the consumer; the decision is made there.

Regime-switching econometrics and POMDP planning. Hamilton’s filter (Hamilton, 1989) carries the full filtered regime probability through time and never conditions on a point estimate of the regime; his recession dates, read off the smoothed probability afterwards, are never fed back into the recursion; POMDP planners (Kaelbling et al., 1998) act on the belief state, not on its argmax. In both, the belief is the carried object and the collapse is the action.

Contemporary agent stacks violate the principle twice. The model thresholds (or the harness argmaxes the returned distribution), and the harness thresholds again on the resulting booleans. Jev’s integration guidance—combine typed probabilities “with deterministic checks in code, then

route” (TypeSafe AI, 2026)—is this pattern stated as best practice. Lemma 2(iii)–(iv) is the analysis of what the second threshold does to the first: (iii) when the second hop consumes the first’s boolean, (iv) when both threshold the same score.

6 Architecture: Belief-State Fuzzy System-One

6.1 Emission block

One shared encoder $\phi(o_t)$ feeds: a regime head $g(o_t) \in \mathcal{S}^K$; per-regime decision heads $h_s(o_t) \in \mathcal{S}^{|D|}$; per-regime membership heads $\mu^{(s)}(o_t) \in [0, 1]^M$ (sigmoid, no softmax); and per-regime type-2 heads emitting Beta parameters $(\alpha_m^{(s)}, \beta_m^{(s)})$ so that $Q_m^{(s)} = \text{Beta}(\alpha, \beta)$ has mean $\mu_m^{(s)}$.

6.2 Transition block

$A \in \mathbb{R}^{K \times K}$ row-stochastic, parameterised by row-wise softmax of a free matrix. We keep the probabilistic reading so that (3.7) is exact Bayes.

6.3 Update block and an open problem

Predict and update by (3.7) with $\Delta b_t(s) = g_s(o_t)/\pi(s)$, where π is the regime marginal under which g was trained, stored with g and never recomputed from a re-estimated A unless g is retrained (Section 3.5). The belief b_t renormalises because it is a probability over a partition; the memberships $\mu_t^{(s)}$ do not, because predicates are not a partition. The belief-weighted mixture $\bar{\mu}_t$ is mathematically well-defined, but whether “membership in M under uncertainty about which regime’s M applies” should be the mixture—for each x , the mixture equals Zadeh’s probability, under b_t , of the fuzzy event on the regime set with membership $s \mapsto \mu^{(s)}(x)$ (Zadeh, 1968)—or a non-additive aggregate such as the minimum over plausible regimes, is a semantic question we state and do not resolve. BSF-S1 sidesteps it by passing the per-regime matrix and letting the actuator choose.

6.4 Propagation and actuator

Pass $O(x_t)$ and, for convenience, b_t itself flagged as *not composable*. The actuator performs the single collapse with context: belief-weighted mixture over regimes, t-norm aggregation across the rule base, regularised centroid defuzzification, cost-sensitive decision, optional sampling.

6.5 Algorithms

Algorithm 1 BSF-S1 step (per observation; $O(K^2 + KM)$ beyond the encoder)

Require: $o_t, b_{t-1} \in \mathcal{S}^K$, A , π , encoder ϕ , heads (g, h, μ, Q)

- 1: $z \leftarrow \phi(o_t)$
- 2: $\Delta b \leftarrow \text{normalise}(g(z)/\pi)$ ▷ scaled likelihood
- 3: $\tilde{b} \leftarrow A^\top b_{t-1}$ ▷ predict, $O(K^2)$
- 4: $b_t \leftarrow \text{normalise}(\tilde{b} \odot \Delta b)$ ▷ update, $O(K)$
- 5: $S_t \leftarrow -\log \sum_s \tilde{b}[s] \Delta b[s]$ ▷ belief surprise
- 6: **for** $s = 1, \dots, K$ **do** ▷ $O(KM)$
- 7: $\mu_t[s, :] \leftarrow \mu^{(s)}(z); \quad Q_t[s, :] \leftarrow (\alpha^{(s)}(z), \beta^{(s)}(z)); \quad P_t[s, :] \leftarrow h_s(z)$
- 8: **end for**

Ensure: $O_t = (\mu_t, Q_t, \Delta b), b_t, P_t, S_t$

Algorithm 2 Actuator: the single collapse

Require: O_t, b_t, P_t ; rule base $\mathcal{R} = \{(\text{ante}_r, C_r)\}$; cost matrix $\text{Cst} \in \mathbb{R}^{|D| \times |D|}$; t-norm T ; floor m_0

- 1: $\bar{\mu} \leftarrow \sum_s b_t[s] \mu_t[s, :]; \quad \bar{P} \leftarrow \sum_s b_t[s] P_t[s, :]$ $\triangleright$ the only marginalisation over s
- 2: **for all** rules r **do**
- 3: $w_r \leftarrow \mathsf{T}(\bar{\mu}[m] : m \in \text{ante}_r)$ $\triangleright$ firing degree; no threshold
- 4: **end for**
- 5: $\mu_{\text{out}}(y) \leftarrow \max_r \min(w_r, C_r(y))$ $\triangleright$ Mamdani aggregation
- 6: $\text{area} \leftarrow \int \mu_{\text{out}}; \quad \text{if } \text{area} < m_0: \mu_{\text{out}} \leftarrow \mu_{\text{out}} + (m_0 - \text{area})/|Y|$ $\triangleright$ regularised centroid
- 7: $u \leftarrow \int y \mu_{\text{out}}(y) dy / \int \mu_{\text{out}}(y) dy$ $\triangleright$ continuous actuator signal
- 8: $d^* \leftarrow \arg \min_d \sum_{d'} \text{Cst}[d, d'] \bar{P}[d']$ $\triangleright$ cost-sensitive decision
- 9: $\bar{Q} \leftarrow \sum_s b_t[s] Q_t[s, :]$ $\triangleright$ kept as a mixture for the consumer

Ensure: $(u, d^*, \bar{Q})$

Algorithm 3 Training

- 1: **Stage 0.** Pretrain ϕ and g on regime labels if available; otherwise choose K by BIC over an HMM fitted on encoder features.
 - 2: **Stage 1 (EM, heads fixed).** Baum–Welch for A with Δb_t as emission terms (scaled-likelihood HMM).
 - 3: **Stage 2 (joint).** The recursion (3.7) is differentiable. Minimise
$$\mathcal{L} = -\sum_t \log \sum_s h_s(o_t)[d_t] b_t(s) + \lambda_\mu \sum_{t,m} \text{CRPS}(Q_{t,m}, \mu_{t,m}^{\text{elicited}}) + \lambda_A \text{KL}(\text{rows of } A \parallel \text{prior}).$$
 $P(o_{1:T})$ is unavailable with scaled likelihoods; the decision NLL through the filter is the computable surrogate.
 - 4: **Stage 3.** Temperature-scale g and each h_s per regime on held-out windows; verify $\text{TCE} \approx 0$ before deployment.
-

Serving cost. Beyond the encoder, Algorithm 1 costs $O(K^2 + KM)$ multiply-adds and Algorithm 2 costs $O(|\mathcal{R}|M + |Y|)$. With $K \leq 16$ and $M \leq 64$ this is under 10^4 operations, orders of magnitude below one encoder pass, with no generative decode. The architecture is system-one compatible by construction.

6.6 Failure-mode inventory

- *Filter misspecification.* Wrong A or g gives exact inference over a wrong model. The belief surprise S_t of Algorithm 1 has a known expectation under the model; a CUSUM on $S_t - \mathbb{E}[S_t]$ raises a drift alarm.
- *K identifiability.* Regimes are latent; K is a modelling choice. Over-specified K degrades gracefully; under-specified K reintroduces Lemma 1 within a merged regime.
- *Membership ground truth.* Elicitation-dependent (Section 8); Q is trained to absorb elicitation disagreement as degree-uncertainty rather than pushing μ toward 0.5.
- *Stale A under drift.* Detected by the surprise monitor; repaired by incremental Stage 1.

7 Experiments

All experiments run on a laptop CPU in under twenty minutes, use only NumPy and SciPy, and are reproduced by one command in the repository. Models are deliberately tiny—logistic heads

on quadratic features and a two-regime filter—because every claim concerns the output object and its composition, not capacity.

7.1 Operational definitions of the two audit metrics

Definition 1 (Trajectory calibration error, TCE_w). Fix a window length w . Every decision model implies a joint predictive distribution for the window error count $N_w = \sum_{t \in \text{window}} E_t$: a memoryless head implies the Poisson-binomial distribution of independent Bernoulli($1 - \max(p_t, 1 - p_t)$) errors, which is the only joint it can express; BSF-S1 implies the distribution obtained by forward-filtering, backward-sampling regime paths (Appendix C) and drawing errors from the per-regime heads. Let F_i be the implied CDF for window i and n_i the realised count. For these discrete predictive distributions we use the randomised probability integral transform $u_i \sim \text{Unif}[F_i(n_i - 1), F_i(n_i)]$, as presented by Czado et al. (2009), who also propose a non-randomised version that could be substituted; checks of PIT uniformity are a cornerstone of calibration assessment (Gneiting et al., 2007). TCE_w reports (a) the dispersion ratio $\hat{\phi} = \text{Var}_{\text{realised}}(N_w) / \text{Var}_{\text{implied}}(N_w)$, where the implied variance is the model’s total variance (within-window variance plus variance of window means), with a window-bootstrap 95% confidence interval, and (b) the Kolmogorov–Smirnov p -value of $\{u_i\}$ against $\text{Unif}[0, 1]$. Only (a) is specific to clustering: any hop-level miscalibration also makes the PIT non-uniform, so (b) is a secondary check to be read together with hop-level ECE, and Section 7.4 includes a time-shuffled control that shows the difference. A model *passes* TCE_w if the confidence interval for $\hat{\phi}$ contains 1. Computing TCE requires realised outcomes for a labelled stream and a fixed decision rule; it does not require model internals.

Definition 2 (Action-mass sensitivity, $\text{AMS}(\varepsilon)$). Fix a perturbation model: *adversarial*, $\delta = \arg \max_{\|\delta\| \leq \varepsilon} |u(x + \delta) - u(x)|$ (for logistic scores, a move of length ε along the weight vector of the gate nearest its threshold), or *Gaussian*, $\delta \sim \mathcal{N}(0, \varepsilon^2 I)$. For a discrete final action a , $\text{AMS}(\varepsilon) = \mathbb{E}_x[\text{TV}(a(x), a(x + \delta))] = P(a(x + \delta) \neq a(x))$, the per-point total variation, equivalently the flip mass. For a continuous actuator signal u , $\text{AMS}(\varepsilon) = \mathbb{E}_x |u(x + \delta) - u(x)|$. The population-level total variation $|\mathbb{E}a(x + \delta) - \mathbb{E}a(x)|$ is *not* used: flips in the two directions cancel and it is near zero for every pipeline. Report $\text{AMS}(\varepsilon)/\varepsilon$ over a grid of ε ; Lemma 2 predicts a constant $\approx \sum_h 2p_h(\tau_h)\rho_h$ for the crisp pipeline ($\rho_h = 1$ for a single gate, as in the first part of E3) and a constant $\leq L_D HL$ for the fuzzy signal. Two rules make the comparison fair: pipelines are compared at *matched action base rate* (a boundary in a low-density region has small flip mass for reasons unrelated to fuzziness), and each pipeline is perturbed along *its own* worst-case direction. Against a closed model the adversarial variant is unavailable (no gradients, no input metric); only the random-perturbation variant can be run, with ε in the units of whatever representation the harness perturbs, and the informative quantity is then the ratio of flip mass between an H -threshold harness and a single-threshold harness *on the same model*, which needs no gradients.

7.2 Generator and protocol

The two-regime stream of Appendix D with $a = 0.01$, $b = 0.04$ ($\pi = (0.8, 0.2)$, $\lambda = 0.95$), emission means differing by 0.5 in each coordinate with unit covariance so that one observation is weakly informative about the regime, and a logistic label link $P(d = 1 \mid o, s) = \sigma(3 o_{k(s)})$ with $k(0) = 1$, $k(1) = 2$ so that the per-regime heads are well specified. The code labels the two regimes 0 and 1 where the lemmas write 1 and 2. E1 and E2 are run on five seeds and reported as mean [min, max]. Both temporal tables include the *exact filter* with the true (A, g, h) : Lemma 1 is a statement about that object, and it is the achievable floor against which every learned system should be read. An ablation keeps BSF-S1’s learned heads and replaces its transition matrix by $A = \mathbf{1}\pi^\top$, regimes redrawn independently every step: whatever BSF-S1 gains over that row

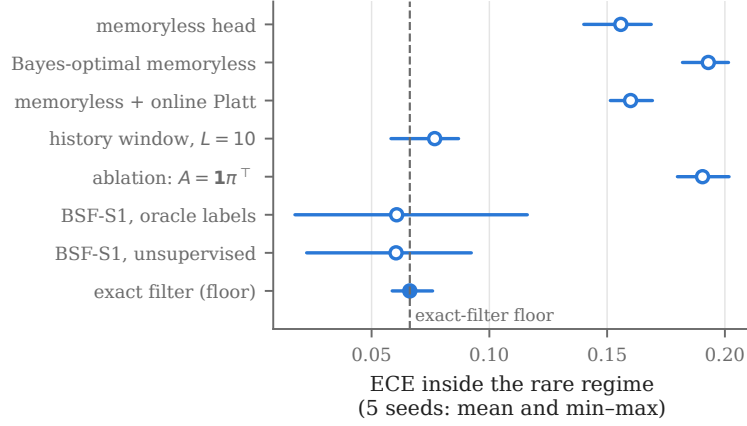


Figure 2: E1, the rare-regime column of Table 2. Memoryless heads cluster far from the exact-filter floor; a better memoryless head is further away; removing the transition matrix from BSF-S1 returns it to the memoryless cluster. The exact filter is not a lower bound on regime-conditional ECE: a learned filter can fall below it on a seed where $\hat{A}$ over-weights the rare regime, at a cost on the other columns.

is attributable to the transition matrix alone. Two BSF-S1 variants are reported: one trained with regime labels (the oracle setting) and one whose regimes are learned without labels by a two-state Gaussian-HMM EM on the observations alone (label agreement 92%), which addresses, for these streams, the oracle-label concern of Section 8.

7.3 E1: Regime-shift stress (Lemma 1(i))

Model	ECE stationary	ECE inside regime 1	ECE shifted, stale A	NLL stationary	Acc. shifted
Memoryless head (quadratic logistic)	0.022 [0.015, 0.025]	0.156 [0.140, 0.169]	0.114 [0.107, 0.121]	0.475 [0.471, 0.482]	0.668 [0.663, 0.675]
Bayes-optimal memoryless head	0.007 [0.005, 0.008]	0.193 [0.182, 0.202]	0.146 [0.140, 0.149]	0.462 [0.458, 0.468]	0.650 [0.648, 0.655]
History-window head, $L = 10$	0.010 [0.009, 0.014]	0.077 [0.058, 0.087]	0.038 [0.028, 0.046]	0.443 [0.440, 0.449]	0.737 [0.729, 0.742]
Memoryless + online Platt, $W = 500$	0.019 [0.017, 0.022]	0.160 [0.151, 0.169]	0.027 [0.019, 0.032]	0.478 [0.475, 0.485]	0.671 [0.668, 0.675]
BSF-S1, oracle regime labels	0.011 [0.008, 0.017]	0.061 [0.017, 0.116]	0.021 [0.010, 0.040]	0.440 [0.425, 0.485]	0.788 [0.758, 0.807]
BSF-S1, unsupervised regimes	0.016 [0.007, 0.027]	0.060 [0.022, 0.092]	0.018 [0.005, 0.038]	0.431 [0.424, 0.442]	0.765 [0.746, 0.780]
Ablation: BSF-S1 heads, $A = \mathbf{1}\pi^\top$	0.007 [0.005, 0.009]	0.191 [0.180, 0.202]	0.144 [0.140, 0.151]	0.462 [0.459, 0.467]	0.651 [0.647, 0.657]
Exact filter, true (A, g, h) (floor)	0.007 [0.005, 0.008]	0.066 [0.059, 0.076]	0.011 [0.008, 0.014]	0.423 [0.420, 0.426]	0.785 [0.781, 0.790]

Table 2: E1, five seeds. Regime-conditional ECE conditions on the true regime, which is outside the filter’s information set, so even the exact filter has a nonzero floor in that column; rows are to be read against it.

Four things the table shows. First, the temporal hole is not a capacity artefact: the Bayes-optimal memoryless head, which has infinite capacity and is perfectly calibrated on the stationary stream (ECE 0.007, the same as the exact filter), is the *worst* row inside the rare regime (0.193) and after the shift (0.146), because a better memoryless head commits more precisely to the frozen prior. Second, the repairs behave as Lemma 1(i) predicts: online recalibration from realised outcomes repairs the shifted-marginal row (0.114 $\rightarrow$ 0.027) and leaves the within-sojourn row untouched (0.160), because it re-estimates one global prior rather than tracking the regime;

a history-window head closes 88% of the gap to the floor inside the rare regime and 73% after the shift, but remains calibrated to a frozen prior. Third, both BSF-S1 variants match the exact-filter floor inside the rare regime and come within a factor of about two of it on the other columns (the unsupervised variant’s stationary ECE is $2.3\times$ the floor), with the unsupervised variant tracking the oracle one throughout; the seed range on the learned filters is wide (regime-1 ECE 0.017–0.116) because Baum–Welch occasionally mis-estimates the transition rates, which is the filter-misspecification failure mode of Section 6 and the reason Algorithm 3 verifies TCE before deployment. Fourth, the ablation isolates the cause: the same learned heads with $A = \mathbf{1}\pi^\top$, π the training marginal, reduce the filter to $b_t = g(o_t)$ and land on the Bayes-optimal memoryless row in every column (rare-regime ECE 0.191 against 0.193, shifted 0.144 against 0.146), so the whole improvement is due to the transition matrix and none to the per-regime heads.

7.4 E2: Trajectory calibration (Lemma 1(ii)–(iv))

Hop-level ECE of the memoryless head is identical on the stream and on its random shuffle on every seed, as Lemma 1(iv) requires. The realised variance of its window error count is 1.83 times the binomial baseline that hop-level calibration implies; equation (3.5) evaluated at the head’s measured regime error rates predicts 1.86. The empirical lag covariance follows $\lambda^k \text{Var}_\pi(e)$ of (3.4) with mean relative L^2 error 0.28 over $k \leq 20$. The learned BSF-S1 passes on

Model	$\hat{\varphi}$ mean [min, max]	passes	KS rejects
Memoryless	1.72 [1.54, 1.95]	0/5	5/5
Bayes-optimal memoryless	2.14 [1.90, 2.41]	0/5	5/5
BSF-S1 (oracle labels)	1.09 [0.88, 1.31]	2/5	1/5
BSF-S1 (unsupervised regimes)	1.15 [0.95, 1.40]	3/5	4/5
Ablation: $A = \mathbf{1}\pi^\top$	2.10 [1.87, 2.31]	0/5	5/5
Exact filter (floor)	0.99 [0.96, 1.03]	5/5	0/5
Control: memoryless, time-shuffled stream	0.94 [0.88, 0.98]	4/5	5/5

Table 3: E2, $w = 50$, five seeds; “passes” counts seeds whose $\hat{\varphi}$ confidence interval contains 1, “KS rejects” seeds with $p < 0.01$. The exact filter passes on every seed; the memoryless heads on none. The time-shuffled control has no clustering by construction: its $\hat{\varphi}$ returns to ≈ 1 while KS still rejects, which is why $\hat{\varphi}$ is the primary statistic and KS the secondary one.

two seeds of five and fails on three: over-dispersed on the two seeds whose estimated switching rate is lowest ($\hat{\varphi} = 1.26$ and 1.31 , $\hat{a} \approx 0.005$ – 0.007 against 0.01) and under-dispersed on one ($\hat{\varphi} = 0.88$, interval $[0.78, 0.98]$). The unsupervised variant passes the dispersion test on three seeds of five but KS rejects on four, consistent with its higher hop-level ECE; the memoryless ablation fails on every seed ($\hat{\varphi}$ 1.87–2.31), like the memoryless heads it reduces to. This is the honesty anchor of Section 8 in numbers: trajectory calibration is a property of exact inference given the model, and a learned model whose A is off in either direction is mis-dispersed by a corresponding amount. It is also why TCE is worth running: it separates the exact filter from its learned approximation, which hop-level ECE cannot (0.007 versus 0.011 in Table 2).

7.5 E3: Cliff cascade (Lemma 2)

Three logistic gates on five-dimensional Gaussian evidence, $\tau = \frac{1}{2}$. *Single gate*, adversarial perturbation: crisp $\text{FM}(\varepsilon)/\varepsilon$ is 0.742, 0.739, 0.777, 0.784 for $\varepsilon \in \{0.005, 0.01, 0.02, 0.05\}$ against the limit $2p(0) = 0.798$; the fuzzy flip mass is exactly 0 and $\text{AMS}/\varepsilon = 0.207$, below the bound $\frac{1}{4}$.

Full pipeline, fair comparison. The crisp conjunction fires on 13.1% of inputs; the product-t-norm-plus-centroid pipeline thresholded at $\frac{1}{2}$ fires on 0.4%, so a naive comparison at $\tau = \frac{1}{2}$ credits fuzziness with a low-density boundary (last column of Table 4). Thresholding the single-collapse pipelines at the base-rate-matched quantile and perturbing each pipeline along its own worst-case

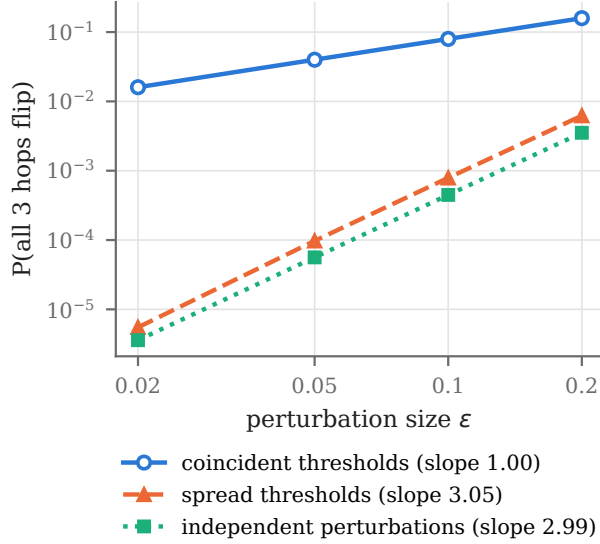


Figure 3: E3, Lemma 2(iv). Probability that all three hops flip under a common-mode perturbation of size ε . Coincident thresholds give slope 1; thresholds spread by a continuous density give slope $H = 3$, the same as independent perturbations.

direction gives the honest numbers: At matched base rate the single-collapse pipeline has about

ε	crisp, own adv.	t-norm + centroid, matched, own adv.	mean of probs, matched, own adv.	crisp, Gaussian	t-norm + centroid, matched, Gaussian	t-norm + centroid at $\frac{1}{2}$, crisp adv. (unfair)
0.005	0.0031	0.0020	0.0022	0.0012	0.0008	0.0001
0.01	0.0062	0.0044	0.0044	0.0025	0.0018	0.0001
0.02	0.0127	0.0090	0.0088	0.0050	0.0036	0.0002
0.05	0.0309	0.0221	0.0215	0.0130	0.0089	0.0005

Table 4: E3 flip mass of the final action, $N = 2 \times 10^5$. The last column is the unfair comparison and is shown only to be disowned.

$0.7\times$ the crisp conjunction’s flip mass under both perturbation models, exactly what Lemma 2(ii) predicts from $\sum_h 2p_h\rho_h$ against $2p_u$ for one boundary; and a plain average of the three scores thresholded once achieves the same. This part of the effect is therefore a *single-collapse* effect, not a t-norm effect, and the paper claims no more for it. The minimum t-norm thresholded at $\frac{1}{2}$ reproduces the crisp conjunction bit for bit $(\min_h \sigma(z_h) \geq \frac{1}{2} \iff z_h \geq 0 \forall h)$: the actuator threshold, not the t-norm, decides whether a cliff exists, which is the content of the principle in Section 5.

Where the t-norm-specific content lies is Lemma 2(iii)–(iv). Cascade: with hop 2 placing weight v on hop 1’s output, $P(\text{gate 2 flips} \mid \text{gate 1 flipped})$ is 0.03, 0.05, 0.14, 0.23, 0.40, 0.48, 0.52 for $v \in \{0.05, 0.1, 0.3, 0.5, 1.0, 1.5, 3.0\}$, approximately $vp(0)$ for small v and saturating near $\frac{1}{2}$; across $\varepsilon \in \{0.005, \dots, 0.05\}$ it stays within 0.388–0.412 at $v = 1$ and 0.022–0.028 at $v = 0.05$, which is the $\Theta(1)$ claim, while the fuzzy hop’s unconditional mean membership change grows linearly, from 0.00020 to 0.00203 at $v = 1$. Coupling, $H = 3$: with $N = 4 \times 10^6$ so that the spread case has 22, 389, 3155, 24873 events at $\varepsilon \in \{0.02, 0.05, 0.1, 0.2\}$, the log-log slope of $P(\text{all flip})$ against ε is 1.00 with coincident crossing points, 3.05 with spread crossing points and 2.99 under independent perturbations, against the predicted 1, H and H (Figure 3).

7.6 E4: Composition algebra (an arithmetic identity)

Rule “ A and B ” evaluated with A entered once and twice. Idempotence gap $\mathbb{E}|\mathsf{T}(a, a, b) - \mathsf{T}(a, b)|$: minimum 0.000, product 0.084; the downstream decision at threshold $\frac{1}{2}$ flips on 6.8% of inputs under product and on none under minimum. Naive-Bayes multiplication of probabilities is numerically the product case; the difference is that it *asserts independence* of two copies of one predicate. This is an identity ($a \cdot a \cdot b \neq a \cdot b$) rather than an experiment; it is reported because the flip rate is the operational cost that harness designers do not usually compute.

7.7 E5: Type-2 separation

Two borderline populations with mean degree $\frac{1}{2}$: sharp, elicited from $\text{Beta}(50, 50)$, and diffuse, from $\text{Beta}(1, 1)$. The point head emits 0.502 and 0.507: the same object. The type-2 head fits $\text{Beta}(49.5, 49.1)$ and $\text{Beta}(1.04, 1.01)$. CRPS on the diffuse population: point head 0.251 (the MAE, $\rightarrow \frac{1}{4}$), type-2 head 0.167 ($\rightarrow \frac{1}{6}$); on the sharp population 0.039 and 0.027. A single number cannot say whether the degree is known; Q can, and is rewarded for it by a proper score. The point head is the baseline that a scalar output type provides, not a strong competitor: any head with a distributional output separates these populations. The claim tested is only that the output *type* must carry a distribution over the degree, and that a proper score rewards it.

7.8 E6: Lemma 1(ii) across regime persistence

E2 tests equation (3.5) at one operating point. E6 sweeps the persistence $\lambda = 1 - a - b$ from 0 to 0.98 with the regime marginal held at $\pi = (0.8, 0.2)$ via $(a, b) = (1 - \lambda)(\pi_2, \pi_1)$, so that the Bayes-optimal memoryless head, which depends only on π , is the same function at every λ and λ is the only thing that varies. Because a memoryless rule satisfies the hypothesis of Lemma 1(ii) exactly, the first columns are a Monte-Carlo check of an identity rather than independent evidence; the informative columns are the exact filter’s. For each λ and three seeds, the memoryless head’s realised variance of the window error count ($w = 50$, 2×10^5 steps) is compared with the binomial baseline and with (3.5), and the exact filter’s own TCE dispersion ratio is computed on a 2.5×10^4 -step prefix. Two things the exact-filter columns show that the

λ	realised / binomial	eq. (3.5)	eq. (3.6), $w \rightarrow \infty$	time-shuffled	exact filter $\hat{\varphi}$	error rate, memoryless / filter
0.00	1.007 [1.001, 1.014]	1.000	1.00	0.981	1.05	0.215 / 0.214
0.50	1.092 [1.078, 1.102]	1.097	1.10	0.991	0.95	0.214 / 0.216
0.80	1.331 [1.314, 1.354]	1.343	1.38	0.990	1.00	0.214 / 0.211
0.90	1.685 [1.613, 1.782]	1.701	1.88	1.002	0.97	0.214 / 0.203
0.95	2.149 [2.109, 2.176]	2.149	2.82	0.997	1.04	0.213 / 0.197
0.98	2.796 [2.769, 2.833]	2.781	5.89	0.991	1.04	0.215 / 0.182

Table 5: E6, Bayes-optimal memoryless head and exact filter. Equation (3.5) predicts the memoryless head’s overdispersion to within 1% at every λ , and the time-shuffled stream stays at 1. The asymptotic factor (3.6) overstates a 50-step window’s excess by a fraction of roughly $(1 - \lambda)^{-1}/w$. The exact filter’s dispersion ratio stays in $[0.89, 1.11]$ and it passes TCE on 18/18 runs.

identity cannot. The filter’s trajectory calibration holds at every persistence level, including $\lambda = 0$, where there is nothing to remember and it coincides with the memoryless head in error rate. And the value of memory grows with persistence: the filter’s error rate falls from 0.214 to 0.182 as λ rises, while the memoryless head’s stays at 0.215.

The closed forms of Lemma 1 were also checked directly at $a = 0.05$, $b = 0.10$, $e = (0.05, 0.40)$, $T = 50$ over 2×10^5 trajectories: $\text{Var}(N_T)$ is 20.30 simulated against 20.31 from (3.5) and 6.94 for the binomial baseline, and $P(N_T = 0)$ is 0.0071 against 0.0001 under independence.

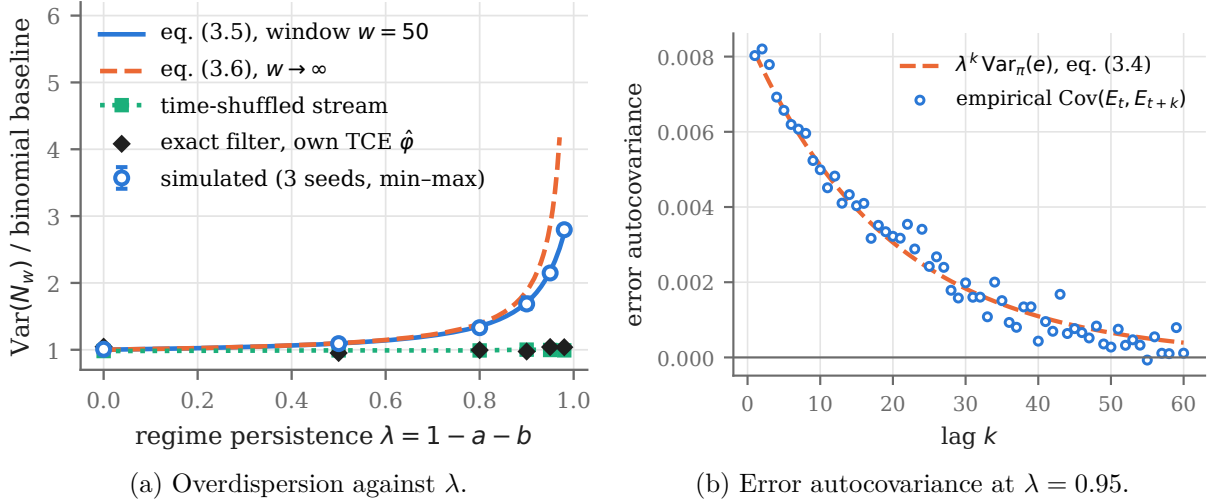


Figure 4: E6, Lemma 1(ii). (a) Bayes-optimal memoryless head: realised variance of the window error count relative to the binomial baseline that hop-level calibration implies, with eq. (3.5) and its limit (3.6); the exact filter’s own dispersion ratio is shown for comparison. (b) Empirical lag autocovariance of the memoryless head’s error indicator against $\lambda^k \text{Var}_\pi(e)$ from eq. (3.4); relative L^2 error 0.10 over $k \leq 60$.

7.9 Ablations and what remains

The memory ablation ($A = \mathbf{1}\pi^\top$) is reported in E1 and E2. The remaining two ablations of the output object are covered by existing experiments rather than new ones: thresholding μ is the crisp pipeline of E3, and collapsing Q to its mean is the point head of E5. Not yet run: a probabilistic classifier head as a stronger competitor to the type-2 head in E5, and experiments on a real decision stream.

8 Threats to Validity and Open Problems

Jev is closed. All statements are about the class of memoryless typed calibrated heads as publicly described; nothing here is an empirical refutation of Jev. Two public facts limit what we can assert. The Score primitive returns a bounded scalar plus a probability per ordered level, so some vagueness can be expressed by schema choice, though over a fixed level set rather than as a type-2 pair. The State object passed to each call may include whatever history the integrator chooses to serialise, so a harness can approximate memory by prompt construction. Neither changes the structural claim: no exact recursion over a latent regime exists inside the model, and no membership-plus-uncertainty pair exists in the output type.

Oracle regime labels. The oracle BSF-S1 rows in Section 7 use regime labels a real deployment would not have. The unsupervised variant (two-state Gaussian-HMM EM on observations alone, 92% label agreement) matches it on E1, which removes the concern for these streams; whether regimes remain identifiable on realistic evidence is the K -identifiability problem of Section 6 and is left open, and in E2 it passes the dispersion test on three seeds of five (oracle variant two) but KS rejects on four of five (oracle variant one), consistent with its higher hop-level ECE.

Membership ground truth. Elicited degrees depend on the elicitation protocol. The type-1/type-2 renormalisation across regimes (Section 6.3) is an open semantic question, stated as such: the architecture is designed so that the question is deferred to the actuator rather than answered implicitly by the model.

Filter misspecification. A wrong (A, g, h) yields exact inference on a wrong model. The surprise monitor is a specification, not a guarantee.

Honesty anchor. We make no claim of escaping the approximation regime. Learning A, g, h, μ, Q is statistical estimation and inherits every failure mode of the underlying network. What changes is the division of labour: *inference*—the recursion (3.7), the t-norm composition, the single defuzzification—is exact given the model. The paper’s claim is that the approximation should live in the parameters, not in the algebra.

9 Discussion

What changes for benchmarking. A benchmark that reports ECE alone certifies $\bar{e}$ and nothing about φ or about action stability. We propose that decision-model benchmarks report TCE and AMS alongside ECE, on streams with a planted regime structure and on harnesses with planted coincident thresholds. Both metrics can be computed from the outside, against a closed model, using only its returned probabilities and a harness (for AMS, the random-perturbation variant). Two models with equal ECE are not equally safe, and a benchmark that cannot tell them apart is not measuring safety.

Industrial precedent as evidence. Fuzzy control, probabilistic forecasting, regime-switching econometrics and POMDP planning are four traditions with largely separate literatures and communities. Each carries the uncollapsed object to the point of action. When separate traditions agree on a design, the agreement is suggestive evidence about the problem rather than about any one tradition; it is not a proof, and Section 7 is where the claim is tested.

Positioning. This is not an anti-neural argument. In BSF-S1 the network does all the perception: it parameterises A, g, h, μ and Q , and none of those objects could be hand-built at the scale of real evidence. What the network should not do is the reasoning that decades-old algebras already do exactly. The synthesis is neuro-probabilistic-fuzzy: networks parameterise, algebras compose, and the actuator collapses.

Open problems. (1) Type-2 filter semantics: the correct combination rule for regime-indexed memberships. (2) Learned parametric t-norms (Frank, Hamacher families) with interpretability preserved. (3) Belief-aware thresholds, τ as a function of b_t , and their interaction with Lemma 2. (4) A theory of trajectory calibration as a proper-scoring object in its own right.

10 Conclusion

Two primitives, two lemmas, one principle, one architecture. System-one decision models are the right idea—software needs an interface it can depend on, and typed calibrated probabilities are a far better interface than strings—but the type signature is not yet finished. Give these models a memory of the world’s hidden state and a language for its borderline cases, carry both to the point of action, and collapse nothing until that moment.

Acknowledgments and AI-usage statement

The author used Claude (Anthropic; the Fable 5.1 and Opus 5.5 models) as a writing and coding assistant: it expanded outline bullets into prose, drafted the experiment code, and typeset the manuscript under the author’s direction. All claims, lemmas, proofs, experimental design, interpretation of results and errors are the author’s sole responsibility. No AI system is an author

of this work. Code, results and the full revision history are at <https://github.com/dnakhoa/jev-deferred-crispification>.

References

- D. Almeida. Introducing System One Models & Jev. TypeSafe AI blog, 15 September 2026. <https://typesafe.ai/blog/introducing-system-one-models-and-jev>
- TypeSafe AI. System One; Primitives (Questions): Choice, Score, Noul. TypeSafe AI documentation, undated, accessed 1 October 2026. <https://docs.typesafe.ai/concepts/system-one>; <https://docs.typesafe.ai/primitives>
- T. Claburn. TypeSafe AI debuts model for machines that plays Doom. *The Register*, 16 September 2026. <https://www.theregister.com/ai-and-ml/2026/09/16/typesafe-ai-debuts-model-for-machines-that-plays-doom/5296711>
- L. A. Zadeh. Fuzzy sets. *Information and Control*, 8(3):338–353, 1965.
- L. A. Zadeh. The concept of a linguistic variable and its application to approximate reasoning—I. *Information Sciences*, 8(3):199–249, 1975. Parts II and III: 8(4):301–357 and 9(1):43–80.
- E. H. Mamdani and S. Assilian. An experiment in linguistic synthesis with a fuzzy logic controller. *International Journal of Man-Machine Studies*, 7(1):1–13, 1975.
- L. R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286, 1989. doi:10.1109/5.18626.
- J. D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989. doi:10.2307/1912559.
- C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *Proceedings of ICML*, PMLR 70:1321–1330, 2017.
- T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- L. P. Kaelbling, M. L. Littman, and A. R. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1–2):99–134, 1998.
- D. Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, 2011.
- H. Bourlard and N. Morgan. *Connectionist Speech Recognition: A Hybrid Approach*. Kluwer Academic Publishers, 1994. ISBN 0-7923-9396-1.
- E. P. Klement, R. Mesiar, and E. Pap. *Triangular Norms*. Trends in Logic, vol. 8. Kluwer Academic Publishers, Dordrecht, 2000.
- J. M. Mendel. Type-2 fuzzy sets and systems: an overview (corrected reprint). *IEEE Computational Intelligence Magazine*, 2(2):20–29, May 2007. Corrected reprint of 2(1):20–29, February 2007.
- M. Saerens, P. Latinne, and C. Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: a simple procedure. *Neural Computation*, 14(1):21–41, 2002. doi:10.1162/089976602753284446.
- Z. Lipton, Y.-X. Wang, and A. Smola. Detecting and correcting for label shift with black box predictors. In *Proceedings of ICML*, PMLR 80:3122–3130, 2018.

- A. P. Dawid. The well-calibrated Bayesian. *Journal of the American Statistical Association*, 77(379):605–610, 1982. doi:10.1080/01621459.1982.10477856.
- D. P. Foster and R. V. Vohra. Asymptotic calibration. *Biometrika*, 85(2):379–390, 1998. doi:10.1093/biomet/85.2.379.
- U. Hébert-Johnson, M. P. Kim, O. Reingold, and G. N. Rothblum. Multicalibration: calibration for the (computationally-identifiable) masses. In *Proceedings of ICML*, PMLR 80:1939–1948, 2018.
- Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. Dillon, B. Lakshminarayanan, and J. Snoek. Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems 32*, 2019.
- I. Gibbs and E. J. Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems 34*, pages 1660–1672, 2021.
- L. A. Zadeh. Probability measures of fuzzy events. *Journal of Mathematical Analysis and Applications*, 23(2):421–427, 1968. doi:10.1016/0022-247X(68)90078-4.
- E. Hüllermeier and W. Waegeman. Aleatoric and epistemic uncertainty in machine learning: an introduction to concepts and methods. *Machine Learning*, 110(3):457–506, 2021. doi:10.1007/s10994-021-05946-3.
- T. Gneiting, F. Balabdaoui, and A. E. Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society: Series B*, 69(2):243–268, 2007. doi:10.1111/j.1467-9868.2007.00587.x.
- C. Czado, T. Gneiting, and L. Held. Predictive model assessment for count data. *Biometrics*, 65(4):1254–1261, 2009. doi:10.1111/j.1541-0420.2009.01191.x.
- S. Chib. Calculating posterior distributions and modal estimates in Markov mixture models. *Journal of Econometrics*, 75(1):79–97, 1996. doi:10.1016/0304-4076(95)01770-4.
- S. Frühwirth-Schnatter. Data augmentation and dynamic linear models. *Journal of Time Series Analysis*, 15(2):183–202, 1994. doi:10.1111/j.1467-9892.1994.tb00184.x.
- C. K. Carter and R. Kohn. On Gibbs sampling for state space models. *Biometrika*, 81(3):541–553, 1994. doi:10.1093/biomet/81.3.541.
- M. Kull, M. Perello Nieto, M. Kängsepp, T. Silva Filho, H. Song, and P. Flach. Beyond temperature scaling: obtaining well-calibrated multi-class probabilities with Dirichlet calibration. In *Advances in Neural Information Processing Systems 32*, 2019.

A Notation

$o_t, d_t \in D, s_t \in \{1, \dots, K\}$	evidence, typed decision target, latent regime
$A, \pi, \lambda, \lambda_i$	transition matrix, its stationary distribution, its non-unit eigenvalues
$g(o), h_s(o)$	regime head $\approx P_{\text{tr}}(s \mid o)$; per-regime decision head $\approx P(d \mid o, s)$
Δb_t	scaled likelihood $\propto g(o_t)/\pi$ (composable; π = training-time marginal)
b_t	filtered belief $P(s_t \mid o_{1:t})$ (not composable)
$\mu \in [0, 1]^{K \times M}, Q$	per-regime membership over M predicates; type-2 distribution over the degree
τ, T, S, D	threshold, t-norm, t-conorm, defuzzifier
$E_t, N_T, \bar{e}, e_s, \varphi$	error indicator, window count, mean and regime error rates, inflation factor

B Full Proofs

Proof of Lemma 1(ii). Let $e = (e_1, e_2)^\top$ and $\Pi = \text{diag}(\pi)$. By the conditional independences, $P(E_t = 1, E_{t+k} = 1) = \sum_{s,s'} \pi(s) (A^k)_{ss'} e_s e_{s'} = e^\top \Pi A^k e$. Every 2×2 row-stochastic A has eigenvalues 1 and $\lambda = 1 - a - b$, with right eigenvector $\mathbf{1}$ and left eigenvector $\pi^\top$ for eigenvalue 1, so $A^k = \mathbf{1}\pi^\top + \lambda^k(I - \mathbf{1}\pi^\top)$. Then $e^\top \Pi \mathbf{1}\pi^\top e = (\pi^\top e)^2 = \bar{e}^2$ and $e^\top \Pi(I - \mathbf{1}\pi^\top)e = \sum_s \pi(s) e_s^2 - \bar{e}^2 = \text{Var}_\pi(e)$. Subtracting $\mathbb{E}[E_t]\mathbb{E}[E_{t+k}] = \bar{e}^2$ gives (3.4). For (3.5), $\text{Var}(N_T) = \sum_t \text{Var}(E_t) + 2 \sum_{t < t'} \text{Cov}(E_t, E_{t'}) = T\bar{e}(1 - \bar{e}) + 2\text{Var}_\pi(e) \sum_{k=1}^{T-1} (T - k)\lambda^k$. Dividing by $T\bar{e}(1 - \bar{e})$ and letting $T \rightarrow \infty$, $\sum_{k \geq 1} \lambda^k = \lambda/(1 - \lambda)$ for $|\lambda| < 1$ gives (3.6). Sojourn in regime 1 is geometric with success probability a and mean $1/a$. For K regimes with A diagonalisable, replace the two-term decomposition by $A^k = \sum_i \lambda_i^k P_i$ with $\lambda_1 = 1$, $P_1 = \mathbf{1}\pi^\top$. $\square$

Proof of Lemma 1(iii). Write $c_s = 1 - e_s$, so $P(N_T = 0) = \mathbb{E}[\prod_t c_{s_t}]$. Every 2×2 stochastic matrix with $\lambda \geq 0$ admits the representation $A = (1 - \lambda)\mathbf{1}\pi^\top + \lambda I$: at each step the chain redraws s from π with probability $1 - \lambda$ and keeps it with probability λ . Conditioning on the redraw times partitions $[1, T]$ into independent frozen blocks of lengths $\ell_1, \dots, \ell_J$. Within a block, $\mathbb{E}[\prod c] = \sum_s \pi(s) c_s^{\ell_j} \geq (\sum_s \pi(s) c_s)^{\ell_j} = (1 - \bar{e})^{\ell_j}$ by Jensen, since $x \mapsto x^\ell$ is convex on $[0, 1]$, with strict inequality iff $\ell_j \geq 2$ and c is non-constant on the support of π . Multiplying over blocks and taking expectation over the partition gives $P(N_T = 0) \geq (1 - \bar{e})^T$, strict when some block has length at least two with positive probability ($\lambda > 0$) and $\text{Var}_\pi(c) > 0$. The same argument with e_s in place of c_s gives $P(N_T = T) = \mathbb{E}[\prod_t e_{s_t}] \geq \bar{e}^T$ under the same conditions, which is the upper tail. $\square$

Proof of Lemma 1(iv). ECE and averaged proper scores are functions of the empirical measure $T^{-1} \sum_t \delta_{(f(o_t), y_t)}$, which is invariant under permutations of t . $\square$

Proof of Lemma 2(i). For the product t-norm, on $[0, 1]^H$,

$$\left| \prod_h a_h - \prod_h b_h \right| = \left| \sum_h \left(\prod_{j < h} b_j \right) (a_h - b_h) \left(\prod_{j > h} a_j \right) \right| \leq \sum_h |a_h - b_h|.$$

For minimum, with $\min a = a_i \leq \min b = b_j$, $\min b - \min a \leq b_i - a_i \leq \max_h |a_h - b_h|$. Combine with the Lipschitz constants of μ_h and D . $\square$

Proof of Lemma 2(ii). Lower bound: for x with $\tau_h - c\varepsilon \leq f_h(x) < \tau_h$, moving distance ε along $\nabla f_h / \|\nabla f_h\|$ raises f_h by at least $c\varepsilon$ (mean value theorem with $\|\nabla f_h\| \geq c$ near the boundary), crossing τ_h ; so, on the part of the band where the remaining gates pass, $x \in F_\varepsilon$, and this set has P_X -mass $\approx \rho_h \int_{\tau_h - c\varepsilon}^{\tau_h} p_h \approx c p_h(\tau_h) \rho_h \varepsilon$. Upper bound: if $|f_h(x) - \tau_h| > L\varepsilon$ for all h then no ε -ball crosses any threshold; the complement has mass at most $\sum_h 2L p_h(\tau_h) \varepsilon$. $\square$

Proof of Lemma 2(iii),(iv). (iii) is (4.2): by the conditional-density assumption the one-sided band of width c' below τ_{h+1} has conditional mass bounded below uniformly in ε , and every x in it crosses the threshold when a_h switches from 0 to 1 (the band above τ_{h+1} handles 1 to 0). (iv): with a common crossing value z^* , the set of z within ε of z^* has P_Z -measure $\approx p_Z(z^*)\varepsilon$, and monotonicity of each ϕ_h in z forces all H thresholds to be crossed together. If the per-hop crossing values $z_h^*(\xi_h)$ have a continuous joint density, all H must fall in a window of width 2ε (measure $\Theta(\varepsilon^{H-1})$) and z must fall in the intersection (measure $\Theta(\varepsilon)$), giving $\Theta(\varepsilon^H)$; under independent per-hop perturbations the flip events are independent and the product of H masses of order ε is again $\Theta(\varepsilon^H)$. $\square$

C Forward-Filter/Backward-Sample for TCE

Forward-filtering backward-sampling draws whole latent paths from their joint posterior; the sampler was proposed at about the same time by [Carter and Kohn \(1994\)](#) and [Frühwirth-Schnatter \(1994\)](#), who named it, and [Chib \(1996\)](#) gives the discrete-state recursion for Markov mixture (Markov-switching) models. Run Algorithm 1 to obtain $b_{1:T}$. Sample $s_T \sim b_T$; for $t = T - 1, \dots, 1$ sample $s_t \propto b_t(s)A_{s,s_{t+1}}$; draw $E_t \sim \text{Bernoulli}(1 - h_{s_t}(o_t)[d_t^*])$; repeat R times to obtain the implied distribution of N_w . The memoryless head has no joint model beyond independence, so its implied distribution is Poisson-binomial with its own p_t .

D Synthetic Generators

```

1: function REGIMESTREAM( $T, a, b, n$ )
2:    $A \leftarrow \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ ;  $\pi \leftarrow (b, a)/(a+b)$ ;  $s_1 \sim \pi$ ;  $s_{t+1} \sim A[s_t, :]$ 
3:    $o_t \sim \mathcal{N}(m_{s_t}, \Sigma_{s_t})$  in  $\mathbb{R}^n$ ;  $d_t \sim \text{Bernoulli}(\sigma(w_{s_t} \cdot o_t + c_{s_t}))$   $\triangleright$  logistic link so heads are well specified
4:   return ( $o, d, s$ )
5: end function
6: function BORDERLINEPOPULATION( $N, \text{sharp}$ )
7:    $o \sim$  uniform on the decision-boundary band;  $(\alpha, \beta) \leftarrow (50, 50)$  if sharp else  $(1, 1)$ 
8:    $\mu_m^{\text{elicited}} \sim \text{Beta}(\alpha, \beta)$  per predicate  $m$ 
9:   return ( $o, \mu^{\text{elicited}}$ )
10: end function

```

E1 and E2 use $\lambda = 0.95$; E6 sweeps $\lambda \in [0, 0.98]$ at fixed π via $(a, b) = (1 - \lambda)(\pi_2, \pi_1)$. Choose (w_s, c_s) so that $P(d \mid o, s)$ differs across s , otherwise $\gamma(o) \equiv 0$ and E1 is vacuous by design.